Cybersecurity & Networking Fundamentals

Minor Project Assessment — Complete Answers (Q1–Q13)

SECTION A — CYBERSECURITY FUNDAMENTALS

Q1. Explain cybersecurity and the CIA Triad.

Cybersecurity is the practice of protecting computers, networks, applications and data from unauthorized access, damage, theft and disruption. It is important because it protects privacy, prevents financial loss, keeps services available and helps organizations maintain trust.
Confidentiality: Only authorized people can view information. Example: a hospital allows only authorized staff to open a patient's record.
Integrity: Information remains accurate and is not changed without permission. Example: a bank protects account balances from unauthorized modification.
Availability: Systems and information are accessible when needed. Example: a company uses backups and redundant servers so its website remains available after a hardware failure.

Q2. Differentiate vulnerability, threat, risk and attack.

• **Vulnerability:** A weakness that can be exploited. Example: an outdated server with an unpatched flaw.

• **Threat:** Something that could cause harm by exploiting a weakness. Example: a cybercriminal trying to exploit that server.

• **Risk:** The likelihood and impact of a threat exploiting a vulnerability. Example: the chance of data theft and the cost to the company if the server is compromised.

• **Attack:** An actual attempt to compromise a system. Example: sending a crafted request to exploit the server vulnerability.

Q3. Explain any five areas of cybersecurity.

• **Network security:** Protects network traffic and devices using firewalls, segmentation and monitoring.

• **Application security:** Finds and fixes software weaknesses through secure coding, testing and updates.

• **Cloud security:** Protects cloud-hosted data, applications and services through access controls, encryption and secure configuration.

• **Endpoint security:** Protects laptops, desktops and phones using patching, anti-malware and endpoint detection tools.

• **Information security:** Protects data against unauthorized access, alteration or loss, using permissions, encryption and backups.

SECTION B — NETWORKING FUNDAMENTALS

Q4. Define a computer network and explain LAN, MAN, WAN and PAN.

A **computer network** is a group of connected devices that communicate and share data or resources. Organizations use networks to share files, printers, applications and internet access, and to support collaboration and centralized management.
LAN (Local Area Network): Covers a small area; example: computers in a college lab.
MAN (Metropolitan Area Network): Connects sites across a city; example: a city-wide network linking a company's offices.
WAN (Wide Area Network): Connects networks across large geographic areas; example: a company network linking offices in different countries. The internet is the largest example of a WAN.
PAN (Personal Area Network): Covers a person's immediate area; example: a phone connected to wireless earbuds using Bluetooth.

Q5. Explain hub, switch, router, firewall and access point.

• **Hub:** Connects devices and repeats incoming data to all ports; it does not intelligently filter traffic.

• **Switch:** Connects devices within a LAN and forwards Ethernet frames to the appropriate port using MAC addresses.

• **Router:** Connects different IP networks and forwards packets using IP addresses; commonly connects a LAN to the internet.

• **Firewall:** Applies security rules to allow or block network traffic.

• **Wireless access point (AP):** Allows Wi-Fi devices to connect to a wired network.

• **Switch vs router:** A switch mainly connects devices within the same LAN, while a router forwards traffic between different networks.

Q6. What happens when a user enters example.com in a browser?

1. The browser checks whether it already knows the site's address. If needed, it asks DNS to translate *example.com* into an IP address.
2. The device checks whether the destination is on its local network. Since a public website normally is not local, it sends packets to its default gateway.
3. The gateway/router forwards packets toward the internet. Routers along the path use IP addresses to move packets between networks.
4. The device establishes a transport connection, commonly TCP for HTTPS, and performs TLS encryption negotiation for secure HTTPS communication.
5. The browser sends an HTTP request to the web server. The server processes it and returns a response, such as HTML, CSS, JavaScript and images.
6. The browser downloads the required resources and renders the webpage. TCP/IP provides the communication framework: IP handles addressing/routing, and TCP (when used) provides reliable ordered delivery.

```
Flow diagram:
User enters example.com
          |
          v
     DNS lookup
 (domain -> server IP)
          |
          v
 Device sends packets
          |
          v
 Default gateway / router
          |
          v
 Internet routers
          |
          v
 Web server (HTTPS request)
          |
          v
 HTTP response / webpage
          |
          v
 Browser displays website
```

SECTION C — IP ADDRESSING

Q7. Differentiate IPv4 and IPv6.

IPv4 uses 32-bit addresses written as four decimal octets, for example **192.168.1.10**. It provides about 4.3 billion total address combinations. **IPv6** uses 128-bit addresses written in hexadecimal groups separated by colons, for example **2001:db8::1** (a documentation example). IPv6 was introduced mainly because IPv4 public addresses became scarce. IPv6 provides a vastly larger address space and supports more scalable addressing. IPv4 commonly uses NAT in private networks; IPv6 can reduce the need for address translation, though firewalls are still important.

Q8. Identify the class and private/public status.

Traditional classful ranges: Class A = first octet 1–126, Class B = 128–191, Class C = 192–223. (Classful addressing is historical; modern networks use CIDR.)
• **10.10.10.5:** Class A; private address (10.0.0.0/8).

• **172.16.5.20:** Class B; private address (172.16.0.0/12, covering 172.16.0.0–172.31.255.255).
• **192.168.1.50:** Class C; private address (192.168.0.0/16).
• **8.8.8.8:** Class A; public address.

Q9. For 192.168.20.45/24, identify IP components and show working.

Given: 192.168.20.45/24. The prefix /24 means the first 24 bits (three octets) are the network portion and the last 8 bits are the host portion. The subnet mask is 255.255.255.0.
Network address: 192.168.20.0 (host bits set to 0).
Host portion: 45 in the last octet (binary 00101101).
Broadcast address: 192.168.20.255 (host bits set to 1).
Usable host range: 192.168.20.1–192.168.20.254.
Usable hosts: 2^(32–24) – 2 = 2^8 – 2 = 254.

SECTION D — SUBNETTING & CALCULATIONS

Q10. For 192.168.10.0/26, calculate the subnet details.

Subnet mask: /26 = 255.255.255.192.
Number of /26 subnets from a /24: 2^(26–24) = 4 subnets.
Host bits: 32–26 = 6. Usable hosts per subnet = 2^6–2 = 62.
Block size: 256–192 = 64 in the last octet. The four subnet networks are .0, .64, .128 and .192.
For the given subnet **192.168.10.0/26**:
• Network address: 192.168.10.0
• First usable: 192.168.10.1
• Last usable: 192.168.10.62
• Broadcast: 192.168.10.63

Q11. Design subnets for IT (50), HR (30), Finance (20), Management (10).

Use Variable Length Subnet Masking (VLSM), allocating the largest network first. The required subnet sizes are: IT needs /26 (62 usable); HR needs /27 (30 usable); Finance needs /27 (30 usable, since /28 has only 14); Management needs /28 (14 usable). These fit inside 192.168.10.0/24 without overlap.

Department	Network / CIDR	Subnet mask	Usable range	Broadcast	Usable hosts
IT (50)	192.168.10.0/26	255.255.255.192	192.168.10.1–62	192.168.10.63	62
HR (30)	192.168.10.64/27	255.255.255.224	192.168.10.65–94	192.168.10.95	30
Finance (20)	192.168.10.96/27	255.255.255.224	192.168.10.97–126	192.168.10.127	30
Management (10)	192.168.10.128/28	255.255.255.240	192.168.10.129–142	192.168.10.143	14

Note: This design supports the stated host requirements. It leaves 192.168.10.144–192.168.10.255 unallocated for future expansion. In practice, reserve addresses for gateways, servers and infrastructure within each subnet.

SECTION E — SCENARIO-BASED QUESTIONS

Q12. Design a basic network for 45 employees, 5 servers and 10 network devices.

Network type: A LAN within the company building, connected to the internet through a WAN/ISP.
Required equipment: ISP modem/ONT, business router, firewall (or router with suitable firewall features), managed Ethernet switches with enough ports and expansion capacity, Wi-Fi access points, Ethernet cabling, 5 servers, employee computers, and a UPS for critical devices. VLANs can separate employees, servers, guests and network management.
Roles: The switch connects local wired devices; the router forwards traffic between internal networks and the internet; the firewall enforces access/security rules and blocks unwanted traffic; access points provide wireless access. Servers should be placed in a controlled server VLAN and access limited by firewall/ACL rules.
Website request: An employee's device resolves the domain through DNS, sends traffic to its default gateway, and the router/firewall forwards allowed traffic to the ISP. Internet routers carry it to the website server. The server responds and

the response returns through the network to the employee's browser. HTTPS encrypts the application traffic.

```
Basic layout:
Internet / ISP
      |
 Modem / ONT
      |
 Router + Firewall
      |
 Managed Core Switch
   |       |        |
Employee  Server   Wi-Fi APs
Switch    VLAN     (employees/guests)
   |
Employee PCs
```

Q13. Troubleshoot a user who can reach local PCs but not external websites.

Check in this order:
1. **Check IP configuration:** Run *ipconfig /all* on Windows. Confirm IP 192.168.1.25, subnet mask 255.255.255.0, a valid default gateway (for example, 192.168.1.1), and DNS server addresses.
2. **Check local connectivity:** Ping another local computer and the default gateway. Since local communication works, the LAN may be functioning, but still verify the gateway and switch/VLAN configuration.
3. **Test an external IP:** Try *ping 8.8.8.8* or use a suitable TCP connectivity test. If an external IP is reachable but websites by name fail, DNS is a likely issue. A failed ping alone is not conclusive because some networks block ICMP.
4. **Test DNS:** Run *nslookup example.com*. If resolution fails, verify DNS server settings and DNS reachability; try an approved alternate DNS server if policy permits.
5. **Check gateway/router:** Confirm the router is powered on, has a working WAN/ISP connection, a valid route/default route, and correct NAT/firewall rules where IPv4 NAT is used.
6. **Check firewall/proxy/VPN:** Ensure security policies, proxy settings, VPN configuration or endpoint software are not blocking web traffic (typically TCP 80/443).
7. **Compare with another device and document results:** If multiple users are affected, investigate the router, DNS service or ISP. If only one user is affected, focus on that device's configuration.

End of answers — Review the subnet calculations and adapt the wording to your instructor's preferred format before submission.